

ActInf MathStream 009.1 ~ Jonathan Gorard: A computational perspective on observation and cognition

Source: [YouTube](#) Playlist: MathStreams Duration: 1:25:24 | Uploaded:

Unknown Transcript method: auto_caption

hello and welcome everyone this is active inference math stream 9.1 on March 5th 2024 we're here with Jonathan gorard and we'll be discussing a variety of topics yet to be determined or are they so thank you for joining and to you for any introduction and we'll really look forward to everyone's comments and questions so thanks again for joining to you okay well yeah thanks very much Daniel for for the the introduction and for inviting me to be here on on active inference it's it's uh I'm looking forward to a very very fun discussion um so uh I don't I mean it's I don't have anything especially prepared to talk about which is probably a good thing because it means we'll we'll be able to extend the kind of the unstructured part of this for for as long as possible but I think just to give a little bit of context I I want to talk about an area where I think some things that I've been working on some collaborators of mine have been working on uh that might have some kind of intersection of Interest with things that you know active inference type people might might care about right so um and in particular that concerns the relationship between kind of computation observation and cognition and specifically using methods that come from category Theory and topos theory and some other kind of branches of of mathematics and theoretical computer science to understand the relationship between uh system specifically the computational and algorithmic complexity of systems versus the computational algorithmic complexity of observers of those systems and how those things trade off between each other so uh so just to give a little bit of context to that I I want to show these are just some visuals from a a paper that I put out uh about a year ago now um and that this kind of really defines This research program that I've been working on for the last year and a half in some form or another which is looking at exactly this trade-off uh using category theoretic Machinery so here's a specification of a touring machine this is just a simple deterministic computation it's saying you know you have a touring machine that has this head State and this tape State and on the next step you're going to replace the the tape

state with something that looks like this the head state with something that looks like that and you're going to scroll the touring Machine Head left or in this case scroll it right uh Etc so this is just a you know specification of a very simple computation a I think this is a a two-state two color touring machine on a simple you know onedimensional tape it's about a simpler computation as you could Define so if you run that thing for some initial condition you'll get uh an evolution that looks like this um and so right now this is just a purely deterministic you know single path Evolution uh but from this we can construct uh we can build a mathematical structure namely we can build a category uh so and the rules for how we build that category are very simple so you know each Arrow here is some some simple computation some application of the touring machine transition function and then what we can do is we can say well anytime we have two arrows that are laid end to end like this we can compose them together to create a third arrow that goes like that I I may even have a picture yes like this so so you know we have a computation F that takes us from X to y a computation G that takes us from y to Z and we can then that we we obtain a composite computation $G \circ F$ that takes us directly from X to Z and we also add some some uh some additional edges some additional arrows uh on each state itself a sort of identity comp an identity operation that Maps a computational state directly to itself and so this combined with uh some axioms of associativity and ident forms a category of Elementary computations so this is a very very simple example but what I want to try and build up towards uh and kind of pump your intuition for is uh a category which I call comp which is a category whose objects are all essentially this the class of all data structures and whose arrows or morphisms are the class of all elementary computations so you start by just applying you know all possible computations or you know in this case for the case of touring machines all possible uh you know touring machine transition functions and then you do this closure operation where you uh you know where where you essentially do what I'm doing here but you you know you you allow those Elementary comp computations to be composed together in arbitrary ways and so that gives you effectively a class of all possible programs so this category comp contains not only all possible data structures as objects but all possible programs as morphisms and this is a a very rich category with some very interesting algebraic structure that we'll kind of again I'm sure we'll allude to in our subsequent discussion um but uh in a sense when we do this when we do this operation of taking what mathematically we call a transitive closure right where where where we allow two Elementary computations to be composed together to produce a third um we are essentially kind of neglecting considerations of computational complexity right

because you know um this Arrow here might correspond to one application of the touring machine transition function this Arrow might correspond to another application of the transition function but this composite Arrow correspond might correspond to two application of the transition function and so somehow when when we when we allow arrows or morphisms to be composed in this way we're neglecting considerations of the complexity of operations so the question then is you know could we imagine uh constructing a generalization of category Theory which takes into account computational complexity so here's a an example of how that would look right so here you can see every Edge every morphism has been tagged with certain computational complexity information in particular it's been tagged with a number specifying what is the minimum number of applications of my transition function what's the minimum number of Elementary computations that I need in order to evolve from this data structure to this data structure so here to go from here to here it's just one to go from here to here it's one to go to from here to here it's one Etc uh but to go to from here to here directly it would be three uh to go from here to here directly it would be two um and you know just for convention we we say that the identity uh the the identity computation the trivial computation always has complexity zero um and then this this so this is a again a fairly simple mathematical structure and you can uh you can construct this uh again using purely category theoretic technology um by building a a particular functor from uh a from the category of computations and from the category of data structures and computations to a what's called a discrete cobordism category and again we might discuss that later on if if people are interested but let me not get too bogged down into the technical details of how we do that but once you've got this it gives us immediately a very nice way of characterizing phenomenal like computational irreducibility so uh there is this idea that um has existed in some form or another since the very early days of theoretical computer science since the days of you know girdle and touring and post and and church and so on uh but was given this term computational irreducibility by by Steven wlr uh with with the where the idea is essentially that uh you just you know intuitively you describe a computation as being irreducible or you know the result of the computation as being irreducibly complex if uh it's not possible to shortcut it in any way right so where you know it takes that computation takes a certain number of steps and there does not exist a shorter computation that would give you the same answer in less time um and one of the nice features of thinking about computations and their complexity algebraically like this is that it gives you a purely algebraic characterization of irreducibility in particular what it says is that irreducible computations are ones for which uh

the computational complexity acts additive purely additively under composition so if it if it's the case that if we compose say two uh um computations of complexity one together uh if the if the resulting composite takes you know has complexity two then it's an irreducible computation if it has complexity less than two like one then that means that we could have we could have jumped directly from the input to the output without having to pass through the two Elementary computations that made it up so that would be an example of a reducible computation so reducible computations are ones whose complexities compose Sub additively in this category theoretic sense um and okay so here's an here's a an illustration of uh showing what what intermediate computational States you had to go through in order to get from one data structure to another data structure so to go from here to here you had to go through steps one to two to go from here to here you had to go through steps one two and uh one two three and four Etc so you can build up a kind of complete algebra of complexity this way which has some some nice properties which again I I I can talk about but I'm I'm let me not get too down in mathematical details right now um but here's the thing I really want to talk about which is what happens when you do when you go to to uh to multi-way systems what happens when you go to non-deterministic computations so now imagine having instead of just a Turing machine with a single rule a single um transition function that just evolves deterministically with a single thread of time now imagine having a Turing machine that has say two transition functions like this one and this one and so at any given point it can apply one of the two um and so now our Evolution instead of just being a single path becomes this kind of branching structure which if if we didn't have any merging would be a tree uh but because we are merging equivalent States it's actually just a kind of more General uh directed graph um and so so it looks like this and this we we we call a multi-way system and so we can build a category out of these multi-way systems as well we can build a category in exact using exactly the same rules so again we we do this transitive closure operation so we we add an edge for every for every um uh every possible composition of these Elementary computations and an identity Edge that Maps every data structure to itself um but it turns out this category has even more structure than the single way uh uh system that that we showed previously because now it's possible to compose computations not just um sequentially in time using ordinary composition but it's possible to compose them in parallel uh across what is sometimes referred to as branching space so so essentially you're you're saying instead of you know the the the the ordinary morphism composition that I showed previously is essentially saying you know I apply this computation I apply this Elementary computation then this Elementary computation

sequentially um whereas this uh parallel composition is saying I apply this computation and this computation in parallel to the same data structure and so that so that parallel operation is what causes these branches right effectively when you when you have two uh you know two threads of time that are branching from the same state like here uh what that's that's arising because we have chosen to apply this Elementary computation and this Elementary computation together in parallel rather than than sequentialized in time um and so this this and so here you can see this this parallelization indicated using what's referred to as a branchial decomposition which is just a kind of a visual way of decomposing what's what's going on between these different threads of time and again there's a purely algebraic characterization of what's going on here which is that uh what we've done is we've taken our our simple category that we started with and we've equipped it with a tensor product structure and so it's become what we fancy call a monoidal category or actually symmetric monoidal category so the tensor so we we now have these two operations we have sequential composition in time and we have this tensor product operation which is a parallel composition in branchial space um and just like we can have just like before we where we equipped uh our our edges our morphisms with certain computational complexity information uh we can do the same thing and we described how those complexities composed sequentially in time we can do the same thing and describe how the complexities compos parallel as one composes morphisms in branchial space and so this allows one in by exactly the same token to quantify multic computational irreducibility rather than just computational irreducibility so now multic computational irreducibility becomes a measure of how additive or subadditive your time complexities are when you compose them in parallel through the tensor product rather than just in sequence through standard morphism composition um okay but I promise I I I am so here's an analogous diagram to the one I showed before showing all the kind of intermediate Med steps uh that that are being uh that that are being applied when one composes uh computation when one constructs computations or indeed multi computations by composing Elementary computations both sequentially in time and in parallel in in branchial space now but I promise I I I am the the point that I'm trying to get to is that it turns out that in addition to just being a useful way to think about computational complexity Theory and to formulate complexity classes like you know polinomial time or nondeterministic polinomial time Etc it turns out this is also an interesting way to think about the role of observation in uh in in sort of computational models of reality because so and here's where I'm going to get a little bit philosophical and I don't have I don't immediately have a slide that or a graphic that I can

show to illustrate this point but so um when we think about modeling a system computationally uh one has to bear in mind that there are really two computations going on right there's there's the computation that the system is itself performing and then there's the computation that the Observer the person who is measuring measuring that system and concluding things from it uh that there's the computation that they that they are performing and somehow you know so when we construct models of reality or when we construct models of systems you know and we want to describe kind of at a meta level what we're doing in computational terms there's our own computation that you know that's going on inside our own internal representation of the world and then there's presumably some external computation that's going on outside and then when we make observations and when we make measurements when we construct theoretical models um what we're doing is we're somehow constructing some kind of encoding function that allows us to take a a concrete physical state of the system we're observing and encode it as some abstract state of the internal model that we have of what's going on and that's all very well but then one but then now we we don't just have one computation to care about we have three right we've got the computation of the system computation of the Observer and the computation this this encoding function computation that's responsible for their for their you know the the interface between their internal model of the world and the external reality and the computational complexities of these computations into play in an extremely interesting way and so the you know part of the reason for trying to develop this algebraic semantics for thinking about computational complexity and multic computational complexity was to try to give one a systematic way to reason about exactly this three-way interplay between systems observers and encoding functions and so in particular when when we make when when an observer makes a model of the world one thing that they're doing is that they are you know for any for any model isn't just you know a complete description of reality there's a certain amount of course graining right there's a certain amount of uh um taking a bunch of states that in the system itself are distinguished but in the internal model are treated as the same they're kind of you know they're cast in the same bucket so in some sense you know how coarse a model is is a measure of how much the encoding function fails to be subjective right and and so so again there's there's a kind of algebraic or c theoretic characterization of what's going on that that you that uh that you know um the the the the fewer of your morphisms are epimorphisms the more coarse your model is you know the more the more abstract or idealized your model of reality is um and so then the interesting thing is that this characterization of multicomp putational irreducibility

irreducibility this measure of how additive or sub additive your complexities are as you compose them together in parallel gives you a measure of the relative complexity of the evolution function that is the the the function that evolves your computation forwards in Time versus the equivalence function that is the function that declares that two computational states two data structures are to be treated as equivalent and that interplay I claim is a kind of abstract meta way of thinking about the interplay between uh the computation of of systems versus the computation of observers because you know so in it say the role of the system is to evolve forwards in time uh whereas the role of the Observer is to take states in the you know in the system that are distinguished in reality and say you know subject to my idealized model I'm going to treat these as the same so the so you know the system is defining the the evolution function but the Observer is defining this equivalence function and so then the the the trade-off in their complexities becomes exactly a trade-off between how you know what are the um uh what are the algebraic rules that describe the complexities as they compose sequentially versus the algebraic rules that describe the complexities as they compose under this tensor product operation um and so yeah I've shown this uh in particular for for for for touring machine systems but this is a very say this is a very general kind of algebraic semantics you can apply it to hypergraphs you can apply to combinators Lambda calculus doesn't matter uh in a sense uh there is just one category up to isomorphism of computation of data structures and computations and there are simply many different ways of parameterizing what that category is doing through things like tring machines or hypergraphs or whatever some you know the the the algebraic formalism transcends the particular details of the computations that one's dealing with um and so yeah as I say what what one ends up with is a is a I think a fairly general formalism for thinking about the interplay between observers and the systems that they observe and uh that gives one a I I promise I'll stop monologuing in a moment and we'll we'll try and pick pick apart what what I'm really talking about here but so I'll just conclude with um you know once once one has that algebraic semantics a whole bunch of things which I think previously would have been at least to me previously seemed like kind of fundamental confus confusions about you know how scientific observation works and and how it interplays with computational models those confusions kind of become much easier to clarify once you think about it in this kind of more compositional way so um to give a very simple example um or a kind of very uh degenerous example um you can you know within this algebraic semantics you can effectively trade off the computational complexity of the system for the computational complexity of the Observer right so um you you you can have um you can have

kind of in effect two degenerate cases you could have the case where the system itself has a completely trivial Evolution function the system itself has has uh you know is doing something completely Elementary in it in in in in how it evolves but then the Observer has some incredibly complicated equivalence function that makes the system look like it's doing something really complicated even though what it's actually doing is something very simple um and so then you so you could have the phenomenon where actually kind of all of the complexity is in the eye of the is in the eye of the Observer you can also have the other degenerate case where the Observer is doing something absolutely trivial where the you know the encoding function or the the observer's own internal representation is just an identity function or something so there's no complexity there but the system is doing something incredibly complex it's doing some some totally s some really sophisticated Universal computation and so that will also appear very complex to that Observer and so and and you can also have any kind of any intermediate you know there's there's this vast interstitial space uh between these two extremes and so one thing that's kind of always uh uh one sort of philosophical problem that I've always kind of uh been interested in ever since I was a kid which is this this sort of um this tension between empiricism versus rationalism right you know the question of uh you know on if you look back at the history of where you know early European philosophy or you know yeah certainly Western you know Western post Enlightenment philosophy you had people like you know decart and livits and and so on who were who or you know and in a more sophisticated way Bishop Barkley with subjective IM materialism who were trying to push for this idea that oh you know all the all the sophistication is what's going on inside the observer's head and you know what goes on in reality is somehow secondary and then you had people like you know Lo and hum and the empiricists who is saying no no we should try and get the Observer as much out of the picture as possible and we should say all the sophistication is going on kind of in the external world and this in one you know one nice consequence of this is it gives one one nice consequence of this formalism is it gives one actually an algebraic way of kind of par amerzing this this spectrum from rationalism to empirism right that that that you can you can choose the the rationalist extreme where you know uh you know you just have some some space of all possible computations and and the Observer is doing all of the work to try to narrow down to a particular one or you can have the kind of empiricist extreme where you know the Observer is a completely Elementary system and you know and and everything and and everything they observe is just being built up from a kind of bottom up uh uh construction all you can have

anything in between and in a sense we we now have I think the beginnings of a mathematical theory that explain that's able to explain how those complexities trade off in a in a very direct way so I think there's potentially places of of of mutual interest there in kind of the in in in thinking about yeah as I say cognition observation measurement scientific modeling and so on in fundamentally computational terms so I think that hopefully that will provide some some context for for for a discussion um thank you great opening there's so many places to uh spin in and jump through I guess I'll start with the two things I wrote down were Unity is plural and at minimum two and beauty is in the eye of the beholder and the way that these kinds of pieces of Timeless wisdom that describe that fundamentally relational component to observers in systems which are not all kinds of systems per se but those kinds of systems those things are true for and then the way in which along the formalism that you described with the system Observer encoding three-way partition and then the way that in free energy principle and the particular physics that interface gets broken out from the agent's perspective into the incoming sensory and the outgoing action so then that results in the four-fold particular partition so maybe just to kind of dwell there there how do we partition the from a category Theory perspective or however the action perception Loop or the engagement Loop like how do we make a topology or compare or contrast different topologies and flows over this kind of seemingly pervasive or Universal interface like concept that's a fascinating question I I so I don't have the answer and this is maybe a place where where um both you Daniel and and perhaps David may may have you know useful perspectives on this because you know I'm I'm you know I I I read K's work a few years back and so I have some familiarity with the terms but I'm by I'm by no means kind of you know an expert on free energy principle or you know active inference and those kinds of things um but I think it's a very good point that you raise and and so you know I I should begin by just being honest and say that you know everything I'm do you know all that I just described is of course an idealization and that you know in reality you know in particular it's an idealization which I think you you were very right Daniel to to kind of pick up on it's an idealization in which we say The Observer is completely kind of non-interacting with the world somehow right that that that you know in a sense that there's just input coming in and nothing nothing coming out uh but of course we know that's not really how observation Works observation is necessarily a kind of two-way process and so what what's needed is not just this kind of very clean algebraic semantics that I've described here which assumes that there's a essentially a one-way function from the world to The Observer but actually uh something more like a kind of second order cybern ICS description

uh of you know what's really going on where where where you have uh you know first order and second order interactions whereas exactly as you say you can get these sort of feedback loops from observation to action and back again which are probably which is I mean still an idealization but probably a more realistic idealization for how for how real observers and real measurement apparatus work so um I just want to begin by saying that I don't know right and and you know the question of how this formalism interplays with things like second order cybernetics and other areas where I know these kinds of questions have been explored explored that's something I I I'm very interested to find out about you know going going forward um but I think H okay so yeah I so at some point may maybe you could help me understand you know potentially where where things might fit in with with the with you know with the kind of broader active active inference um framework um I'm not sure I necessarily have that much more to to comment on than that um yeah other than to say that that you know in a sense okay so maybe you know one one further comment is that um because I mean you you asked specifically about how the the kind of compositional category theoretic perspective might be useful um so I I don't think category theory in itself is going to be the complete answer I think it will be category Theory augmented with some other things computational complexity probably second order cybernetics uh and some other things that I that I may not be aware of um but one place where I think that that Viewpoint is useful at least on a philosophical level is the idea that is the idea that comes about that that that you really obtain by studying mathematical structures in a category theoretic way which is that the identity of something you can Define it both in terms of its intrinsic Properties or you can Define it in terms of you know the stuff that you can do to it right so you know this was really the transition that happened in the foundations of mathematics as a result of people like Samuel lenberg and Saunders mlan um so you know category theory has its origins in this sort of slightly abstruse branch of algebraic topology uh it was you know initially developed by people like Alexander gend deque and John Pier for doing um homological algebra for you know for reasoning about sort of the algebraic structure of topological spaces but then later in the I think 1960s 1970s these two American mathematicians eidenberg and mlan realized that it was useful not just for thinking about topology but for thinking about kind of mathematical structure in general and then later on applied category theorists started saying well maybe it's useful for just thinking about structure in general but the you know the key kind of conceptual or philosophical shift that it imposes is you know historically thanks to the work of people like Kantor and frer and Russell and so on people have thought about mathematical structures in

the foundations of mathematics as you know in terms of set theory and the idea in set theory is you have this you know things like the axiom of extension that effectively say a set is defined by what's inside it right so in other words you know you you you a mathematical structure obtains its identity by you know you break it apart and you look at what's inside um in category Theory it's a completely different view the view instead is you say well no you can't look inside you know it's a fundamental rule of category theory that you can't look inside an object uh you know its internal structure if it has any is sort of out of bounds to you and instead you you give that object identity in terms of how it relates to other objects of the same type right so in other words you know you can ask you know what what can I do to this what what functions can I apply to it what functions can I apply to something else that map into this uh so you know if I want to Define I don't know uh the real numbers or the integers or something in the set theory St you know from a set theory perspective he would say the the you know the the essence of the real numbers are all the numbers that are inside that set all the numbers that are inside $\mathbb{R}$ whereas the category Theory perspective is no the essence of the real numbers are all the functions that you can Define that take real numbers to some other number system or real numbers to themselves or that take some other number system into the real numbers or Etc and you know some of the deepest results category Theory like the on Lemma and other things are telling one in some very precise sense that these two perspectives are really the same at some fundamental level that that you know that identifying an object based on its internal structure based on breaking it apart and asking what's inside and identifying an object by asking what can I do to it and what you know what can this object be transformed into and what things can be transformed into this object those give you exactly the same information it's far from obvious that that's true you know the nmma is a very kind of uh it's one of those results where you can never quite work out if it's obvious or if it's incredibly you know mysterious uh but I would I tend to fall on the side that it's incredibly mysterious it's far from evident that that those two perspectives would really be the same and yet the point you're making Daniel I think is that uh is that in a sense our our historical ways of thinking about scientific observation have tended towards the set theoretic Viewpoint the tended towards the Viewpoint that we we understand systems based on kind of breaking them apart into constituent components um but it perhaps a more realistic view is something more like the category Theory perspective where we say no I I I understand a system by interacting with it right by by by asking what can I do to it and how does it behave as I when I when I perform certain operations to it and that's a fundamentally uh you know that's a fundamentally

two-way process that involves not just passive observation but also kind of active participation and somehow we need to develop a formalism that that kind of incorporates those two elements and maybe you know maybe it already exists and it and it exists in this large literature tree uh of which I'm largely unaware that's partly why I want to be here to to try and find out you know what things I missed so to speak excellent all right a few points self-evident is far from evident also I tend to the mysterious which is to say saying more with less especially for these Frameworks because they're less opinionated so that their space of internal semantics can be larger and then that description that you provided with the relationship between the set and the C category Theory so I kind of summarized it as set is to essential inclusion as category is to relational function now if our concept of organismality or of action in The Niche is constructive compositional material then we are looking for like what is in or out is the microbiome in is the pheromone and the ant colony in or out of that thing because it's looking for like a static material IAL answer and then in contrast the the other side of that coin highlights the the dynamic like whatever it is that's self-organizing of the tornado is the tornado whatever it is that's self-organizing for the ant is the ant um and then also this like hint SL mobus strip or something that those two in the moment are indistinguishable and yet systems that we choose to Define one way or another or keeping both open those design decisions do make all the difference even if for real systems as they're observed there's indistinguishability right right I think that's a that's a very important point and and one which I mean this is a it's it's always a kind of concern I have whenever I start thinking about you know embodied cognition or or you know um Extended phenotype type ideas right that that in a s you know if what one is trying to do here is construct a kind of formalistic model of of observation or of cognition or something then as a kind of first order approximation one has to start by somehow decomposing the world into observers and systems but of course that you know we know that that decomposition is is somehow arbitrarily imposed right and that you know if you take these things to their extremes and you you know you allow essentially everything that the that the agent is interacting with to be consider you know like like not just the microb as you say but also you know tools that they construct or environments in which they exist and so on if you if you as you start to consider all of that to be a component of that organism you know of that agent's phenotype which is a completely reasonable thing to do then uh and and you start to you know you start to say okay well their cognitive processes are not just localized to their brain or their spinal column but are kind of somehow extended to you know the computers they use the paper they write on uh the books they read Etc again

perfectly reasonable thing to do and sort of somehow more uh more descriptive of what's really going on but my fear is always if you take that too far then you know you end up destroying the whole assumption that the uh that the idealization was based on which is that you can neatly decompose you know uh the world into into observers and systems and so I always get a bit nervous when when thinking about that that it's that it's like yes you know in a sense yeah we know this is an approximation and we know that that approximation is not really true but how how you know um how much can can you afford to to sort of uh to loosen your grip on that approximation before the whole thing just kind of falls apart I I I don't really know the answer to that question um but I think it's an interesting one yeah how about ask some questions from chat and then give your first thoughts and then we'll see maybe where that kind of lands with further questions or how it connect to active inference so that sounds good oh by the way should I keep my screen share on or should I should I yeah we might want to go to a figure so it's fine okay yeah all right Quantum Bell wrote how does this help us reason about causality that's a fascinating question okay so um that that's that's another major aspect of of you know why I think this research program is exciting because so um and again this is something where I'm interested to get the kind of active inference perspective because I know this again it's a topic which much has been written and and I I'm I'm largely ignorant um but yes so one question you could ask is uh yeah if if you have a description of a computation like this like uh let's go back up to the touring machine case the the single way touring machine case that's relatively easy to analyze although still far from obvious what's going on so suppose you have a computation of this kind and you want to ask uh what is its causal structure in other words you know for each Edge each time I'm applying this turing machine transition function can I construct some kind of graph you know some some directed graph representation that tells me how these events are linked together so in the uh within the original resarch program this this the so-call WF for model research program that kind of started a lot of these investigations we were looking at this all the time right we were looking at kind of you know taking computations and and looking at their causal structure and trying to you know infer things about what was going on about the you know the semantics of the computation based on uh clausal relationships and at a certain point I started to realize and I think other people had realized this before I did but I'm often slow to to pick these things up I I I and other people started to realize that the notion of causality we were using was kind of nonsense I mean it it was not completely hopeless but it wasn't really causality or it couldn't really be called causality in any in any definite sense so um what do I mean by that so first of

all it was a very technical problem so uh if you're looking at something like a turning machine Evolution or a hypography writing system as we were um then there's a very tempting and OB apparently obvious natural definition of causality that you could use which is to ask you know when you split the world up into events that take you know some part of your data structure as input and output you know some other part of a data structure is output then you can very easily ask well does the output of one event intersect with the input of another event so uh if I show the hypergraph example it's perhaps easier to see so you have a hypography writing rule that looks like this right so you know you have you say if I have a piece of hypergraph that looks like that I replace it with another piece of hypergraph that looks like this so at each time you apply an event you can think of that event as you know ingesting certain hyperedges and kind of you know replacing them with others so you can divide it up into a sort of the the the input hyperedges that are being ingested versus the output hyperedges that are being produced and so then you can ask well do the output did I use did I subsequently ingest in some future event hyperedges that were output in some previous event well if the answer is yes then pretty obviously that future event couldn't have occurred unless the previous event had already occurred so then you could say well then one of those events causes the other so in general you could say that two you know event A causes event B if it's the case that the output uh that the collection of tokens that was produced in the output of event A has a non-zero intersection with the collection of tokens that were ingested as part of the input of event B and that's is a very tempting very natural definition of kind of causality in these systems turns out it doesn't really work I mean it works pretty well but there are cases in which it fails and it fails pretty spectacularly and uh so the kind of canonical case where it fails spectacularly is that you can have events that don't actually do anything right you can have events that just kind of touch an edge touch a token and and output it again unchanged but maybe you know change the it modifies the name it modifies the identifier um but it doesn't actually change anything about the structure of the hypergraph or the touring machine state or whatever so pretty obviously that event doesn't matter it can't it shouldn't be causally related to anything in the future but because it modified because it ingested The Edge and then didn't do anything you know did some identity operation but then you know produced it in the output again it will kind of register as being causally related to any future event that used that edge even though it didn't make any difference um that's just one very obvious example there are other cases where it became clear that whatever this thing was whatever this algorithm was detecting it wasn't really causality so I tried to think about you know what's a

more sensible definition of causality and I I I started working on things to do with you know a slightly kind of blockchain inspired ideas where you say okay well rather than just arbitrarily assigning uh you know identifiers to these tokens uh every time they're created what if I recursively construct the identifier of the token based on its causal history so in other words each token like each hyper Edge or each state in my each um each tape uh Square on my touring machine uh tape I the identifier is not just some random number that gets generated by my algorithm but instead its identifier is a directed graph representation of its complete causal history well then kind of recursively its identifier can only change if the causal history was updated and so you don't end up with these kind of spous causal relations that I described before so that seemed like one tempting uh way of resolving this problem but then I realized actually there's a much more fundamental problem there's a pro there's a philosophical problem with the way that we're thinking about causality which is that it's not really you know so okay this is a long tangent which I'll I'll I'll talk about a little bit but I won't get into the complete details unless people are interested but I ended up I ended up talking to a bunch of philosophers who who you know who worked on causality and people who worked on Parallel programming and Quantum information the and other places where causality was was was studied and asked them kind of basically what what is what what do you mean by CA what is causality what is this thing we're trying to to to Define and in some form or another all of the definitions boiled down to you know event A causes event B if had event A not occurred then event B would not have occurred so in other words you need a counterfactual you need some possible history some possible World in which event A didn't happen but if you're reasoning about a purely deterministic event system like a touring machine that doesn't make any sense because if you're you know if if you have a single touring machine transition function there is no possible World in which that transition function didn't fire in that particular way because if it didn't fire in that particular way you would not be reasoning about that touring machine anymore you'd be reasoning about a different touring machine so suddenly this you know to make sense of these Notions of causality you need a kind of Litan you know modalities view of reality that just doesn't exist for these deterministic computational systems so either you need to Define computation you need to Define causality only at the multi-way level only at the level where you have many computations or possibly all computations happening in parallel and then you can Define causality relative to all of you know relative between them or you were kind of hosed right there wasn't really you know that that seemed like the only kind of the only get out or you need some fundamentally new

philosophical theory of causality that I was not qualified to produce um and so so that that's Again part of the reason what with the part of what motivated this General search program which is trying to think about this category of not just a single computation uh and with a single sequence of data structures because it's clear that you can't in a philosophically meaningful way assign causality in that case but rather uh you know looking at the the the algebraic structure of the category of all possible computations and all possible data structures and in that situation there is a notion of causality you can equip that with and uh there's a nice again a nice mathematical description in terms of of in terms of week two categories and so on which again I can I can talk about if people are interested um but but yeah so it's clear these things are are very deeply related that this this sort of theory of the the category of computations and data structures and the theory of how you assign causality in a meaningful way uh are very deeply related and um I'll just mention one other thing on that topic which is again just area which I I I find quite exciting because it's it's an unexpected spin out of this program which is that so once you have a way of consistently applying causality at a per token level in these systems it gives you way of vastly generalizing what computation is um and you can in particular you can get you can derive something which I call uh well I'm provisionally calling covariant computation although it should probably have a better name than that which is so you know in our traditional kind of touring Church type po you know type models of computation computation is a purely Fords in time operation you know so at every point you know you have a complete data structure and computation is about deriving what is the next state of that data structure so sense it's it's it's only a forwards and time thing you might be able to kind of reconstruct the initial conditions based on some subsequent data structur so you might be able to go backwards in time but that's essentially what you're doing but then you could imagine okay suppose I don't know the complete state of my data structure I know instead I know one part of my data structure but I know it I know its history throughout all of time so you could imagine say an elementary cellular automaton or a touring machine tape where you know nothing about the tape but you know the state of one cell and you know it you know throughout all of time and then the question is what can you infer about the rest of the computation and it turns out that for those kind of structured array type systems you can infer a lot you can actually evolve the the system not forwards in time but sideways in space and obtain a kind of causal Diamond that so okay the the the the the top left top right bottom left bottom right Corners are are undetermined but everything inside that diamond can be determined just from that one row uh or that one column that you know uh you know sort of

extended throughout time and so uh you know and that's a fundamentally different notion of computation so it's a version of computation which is not but you can also have version of computation that are sideways in Branch your space where where you you know uh you know one complete State you you know you know one branch of the multi-way system extended throughout time and then the question is what else can you infer about the you know about the rest of the multi-way system just from that one branch and again the answer turns out to be you can infer a lot but not everything and so just like in the reason I call this covariant computation is because it's very analogous to what happens in relativity so in relativity once you buy into this notion of general covariance and the notion that space and time are kind of fundamentally the same thing then you have to somehow relax your traditional view of what dynamical systems do which is you know we typically think of systems as evolving you know you have a snapshot of your inti of your data at you know localized on a you know for a particular state of space you know on a particular space like hypersurface and then your laws of physics tell you how that space like hypersurface evolves forwards or backwards in time but uh in a covariant picture of physics then you you must also allow for you know your initial data to be defined on a time likee hypersurface and you for you to be able to to evolve that Tim likee hypersurface sideways in space or mixtures of the two and so on and so it's clear that there's a very general a vast generalization of ordinary computation theory that you can def that you can construct that's kind of physics inspired in that sense um in which you can have mixing of space time and kind of multi-way directions um in a in a completely consistent way but to to make those things consistent you need to have a a definite way of assigning causality you need because you know any computation that you do even if it permutes the directions of space and time and branchial space and so on must always somehow uh preserve the causal structure has to respect the causal structure of of what's going on or else it's in consistent and so uh this this question of how you construct a covariant theory of computation is it turns out intimately related to uh to the question of how you take this structures and equip it with a with a consistent notion of causality so very interesting question we could talk about that at Great length but okay to follow with a few pieces it's very related to Professor Mike Len's notion of poly Computing and about the necessity for a causality concept to be created or deployed when the question arises was that me was that action or change due to me um also connectivities even just in the neuroimaging setting which is kind of the Cradle from which active inference and free energy principle AR from it's really important to distinguish the functional effective and anatomical connectivities and that was one of the points that Toby St

Clair Smith made in his dissertation which is that a lot of times the basian graphs don't convey all of the necessary and sufficient information to make the reproducible computation which is one of those kind of what's missing from the graph is what motivated a lot of the category Theory developments in active inference as well as some of the formal ontological works with sumo and Dave here and Adam peace because implementing modal and higher order Logics is really important if it's a possible situation where a mind can have a perspective on a mind and all these things like that um then the ant Turing tape the tape is the pheromone and then the decision space is the nestmates scrolling so when you had a deterministic turing tape that was like a movie because the nestmate couldn't make any choices except for internal action which is kind of side topic but couldn't make any choices on the tape whereas when there's a multi-way which is basically in active inference what we talk about in terms of affordances and the policy space and the temporal depth of planning counterfactuals on action and action conditioned World transition States like the B Matrix all those kinds of topics come into play because if you want to have a causal buffer or grasp on what is it that something that could do otherwise does what does it cause to do when it does or doesn't do otherwise you need something like a deterministic handle around what could be a probabilistic or deterministic but at least multi-way map of some kind of cognitive territory that's a very very interesting perspective I mean so it's again I'm I'm I'm betraying my my ignorance of active inference Theory here but you know it sounds almost like so these these um when you have this kind of interplay between I don't know the you know between sort of EP you know epistemic versus uh pragmatic this you know there there sort of two aspects of of um of of you know of how how this kind of speculative part of cognition Works um I this is something which I I've thought about I mean I I thought about in a completely different context in relation to things like Quantum information Theory but I I wonder if it's I wonder if there's a a potential overlap there between so there are certain situations when thinking about these kinds of systems purely abstractly where you kind of need two different Notions of causality you need a kind of speculative notion that can be that's Dynamic that can be Rewritten and then you need a kind of definite notion that's that's immutable right and so you know a classic example of this is you know say for some something like Quantum information Theory right so you can have superpositions of causal orders you can have Quantum switches you can have uh you know you can have causal structure that exists in uh in in in in superpositions of different kind of directed graph States but then once you apply her Mission operator once you apply a measurement the causal structure is definite because then everything is relativistic and and you

have covariance and you have similar things as I understand with with um with uh with distributed computing with parallel Computing where you you you potentially allow for speculative execution for certain number of steps where you're kind of treeing out this multi-way system and you have a superposition or you know at least a collection of possible causal histories but then eventually you have to choose an actual operation to do and then the causal history you know you have this big block that gets laid down and you know and and then the causal history is is is somehow definite and I I wonder if there's a there's a way of thinking about um uh sort of yeah you know um speculative execution of agents and the interrelation between that speculative execution and you know and agent actions in terms of again this interplay between you know between two different causal structures between a dynamic one versus an immutable one yeah well one funny way to think about that is like a single agent that has this counterfactual contemplative ability could be in the center Place foraging Arena and then imagining with discrete um branching paths like a chess algorithm or like a probability distribution could be like imagining where it could go but not all C of things are the kind of things that make plans of their own actions whereas like an ant colony has nestmates on the ground so they're actually realizing in these finite trajectories the real exoskeleton on the ground that plays out ending up with those simulated trajectories could have been simulated or could have been probabilistically blurred but that's kind of the difference between like the embodiment and like the body moving there for a mammal for an animal and then like the mind simulating it and then just to the um the epistemic and pragmatic trade-off in decision-making so let's just say that we're in that multi-way moment we'll just have two options two different slices of the B variable and the policy selection question is about which way are you going to go which affordance in the moment policy is basically the affordances for the time Horizon of planning but if it's only one time step or just an x one then the afford in space is just the actions that can be taken um one way to make it so that what happens is the likeliest thing path of least action which is kind of what opens up the whole physics of cognitive systems angle in contrast to like a reinforcement or reward learning perspective what makes it the likeliest thing is starting with habit so it could just be drawn from a fixed habitual distribution however for adaptive action habit gets upweighted with expected free energy which is a functional that takes in the policy space which is summing up to one because there a probability over actions and then upweighting policies according to their score on expected free energy which is consisting of epistemic plus pragmatic value so how much is it going to align the observations to be what I like to see that's pragmatic value with a preference what is

my expected Information Gain that's the epistemic value so how those are parameterized make the agent that always seeks out new information or always goes with habit all there's so much policy space because the knobs are not just simple sliders or there are multiple knobs even though they are seemingly quite consilient and minimal like it's hard to imagine less yet especially when there's richness in the environment even simple systems can have like enormously complex or adaptive behaviors I'll just leave it there no I I never really thought of the so okay the yeah two things right so so first of all the the the perspective of you know thinking of an ant colony or a termite Colony or something as being you know akin to a mind that's you know I I was familiar with that perspective from people like Dan Dennett and so on but the the the idea that the individual ants in that colony are in a sense enact they're kind of the hardware enacting the speculative execution that's a very interesting idea it's not speculative for them well yeah no exactly but it's sort of from the mind's perspective I guess it's almost like speculative execution but but it's speculative execution that's being actuated in the physical world um which is which is very interesting I I I not really thought about that before um but then yeah okay so so so then then the yeah the the point you're making about connection to free energy and and and of habit for and so on that okay so um I wonder if so you know if if we're thinking about uh a a you know a model of cognition in which there are these two different you know two two distinct causality Notions the the immutable versus the dynamic one um I wonder if the so you you gave a very very nice account of how you know habit formation sort of works in these in these kinds of formalisms based on you know prior experience of of expected you know free energy so I I I wonder if there's a way of describing that abstractly in terms of something like you know you you you perform the speculative execution step um where you you know you're treeing out several multi-way possibilities uh and initially you kind of you know you you you know nothing or you have no habits you're just kind of you're treeing everything out with kind of equal waiting but then you know for each possible path you're calculating either an actual or an expected free energy and then somehow you know in future speculative executions you wait those paths which you previously had found to have higher free energy as higher um and so you're you know you're more likely to explore those and less likely to explore ones which are you know which have that lower expected value um it feels like something something like that should should fit very very nicely into an algebraic semantics like this um which would be interesting oh how about more questions from the chat okay upcycle Club writes acknowledging the limitations of traditional entropy in multic computations motivates us to develop context specific entropy

metrics can you share some insights towards such efforts yeah uh I I can certainly try um so so yes I mean the first point is that you know it's I think it was there's that famous conversation between John Von nman and Claude Shannon where I think Von nman famously said that like Shannon should call his measure entropy because no one knows what it means right and I I I submit that the reason no one knows what entropy means is because it's dependent I mean okay one of the reasons no one knows what entropy means is because it's dependent on exactly what we've been talking about it's dependent on the equivalence function of the Observer so um it's one of these things like um I don't know maybe this is a stupid analogy to use but it's sorry I'm I'm I'm going to go off on a tangent but I promise it's sort of relevant but so um one thing okay one thing that always breaks my brain is when I try and think about like actuaries and life insurance policies because it's one of those areas where those models only make sense if they're not perfect in a sense right like so if if you had an actuary who knew exactly how long everyone was going to live um and and somehow that information was kind of openly available there would be no like life insurance policies would be pointless it it's but you know whereas also if you had a model that was completely hopeless of predicting how long people would live they would also be hopeless they would you know life insurance policies would also be pointless that somehow the very existence of Actuarial science relies on your model neither being perfect nor being awful it it somehow has to exist somewhere in between um and I say that's something which I I it's one of those topics where if I think about it for too long it all just stops making sense and entropy has very much that same character right because if you if you were lasso demon if you know if you had perfect information about the system that you were observing there's no notion of entropy right it's just every you know every you know every micro State um so that you know so so you know the boltzman formula gives you an entropy value of zero where but you know the notion of entropy only exists once you take once you take that that perfect knowledge of a system and you Co grain it you define as I was describing earlier you introduce an encoding function that is not 100% sub subjective so that now you are mapping certain distinct micro States onto the same coar grained macro State and then now you can ask okay what's the number of micro states that you know how Co is my cor scening what's the number of micro States consistent with this macro State what's the number of different values of my domain that gets mapped to a single point in my codomain of my encoding function and that's what entropy is and so it's it's very closely I mean it is effectively a measure of how good is my course graining so if you had perfect knowledge there's no entropy if you have no knowledge there's no entropy it relies upon

having a not completely trivial but also not 100% you know a slightly subjective but not 100% subjective encoding function um just like with uh life insurance policies but so the reason the reason I'm stating that is because so now it it kind of I think from that perspective becomes a little bit clearer why there are all these different Notions of entropy and why as the questioner was alluding to why entropy seems to be so as a concept seems to be so domain and and system specific because every different system every different Observer will in principle have a different set of of encoding functions a different set of equivalence functions and each one will give rise to a different calculation of entropy um and so one way that you can think about this program this program to try to uh understand the algebraic interplay between time complexity versus kind of equivalence in complexity or or you know computational irreducibility versus versus multic computational reducibility in some sense um that is a program to try to understand how different definitions of entropy relate to each other in these kinds of systems um how you know if I take one idealized Observer that has this equivalence function and I and I ask uh okay suppose now they communicate with this different Observer with a different equivalence in function they come to different understandings of what the entropy of the system is but what is the relationship between their measured entropy values there clearly is one that depends algebraically on some details of the of the distinction between their respective equivalence functions but there doesn't seem to be any yet any general theory for for you know for how those things are related and you know that that's part of the kind of um the of this of this research program um but yeah I mean okay so uh one thing that I will comment on uh although this is a little bit more speculative it's again sort of Quasi philosophical comment but so one place where these Notions of entropy become so one place where the fact that you have all these different Notions of entropy becomes kind of interesting is in fundamental physics so when you start to think about if you try to model physics and and the universe in in these fundamentally computational terms then uh one fairly generic sort of conclusion that you can reach is that gravitation general relativity is essentially an entropic phenomenon um I mean valid and people have kind of talked about this in in in non-computational contexts too but it's very very natural if you start to think about space like hypersurfaces being sort of hypergraphs then you know in order to obtain a Continuum geometry that's compatible with the Einstein equations you need to have certain ergodicity uh you know um you need to be able to make certain ergodicity assumptions on the rewriting which in turn sort of implies certain lower bounds on the entropy of the system so somehow gravitation general relativity is a is a coarse grain Theory you obtain in the limit as the

entropy goes to Infinity but something that's interesting is that quantum mechanics on the other hand is an IDE is an idealization that you obtain in the limit as entropy goes to zero because in in in kind of one of these purely computational models of physics uh the quantum mechanical state of a system is described in terms of its multi-way structure it's described in terms of you know when I have a a kind of a branching program like this let me find one of these like this then you know I can divide it up into these into these sort of um into these simultaneity surfaces and if I associate each each state of the program as being like the analog of a Quantum igen State and the kind of path weightings as being you know analogous to the amplitudes associated with the igen states I can quickly build up a description of this multi-way system in terms of the evolution of some discrete analog of the shredding equation and it turns out you get a theory that is kind of equivalent that is mathematically isomorphic to standard quantum mechanics out of it so uh quantum mechanics is sort of you know inextricably bound up with with with with the with the with the phenomenon of the multiway system but if you take the entropy to Infinity here then you're effectively uh then the then the sophistication of your of your equivalence function becomes arbitrarily large which means that you can describe any essentially any pair of States as being equivalent um and so uh it turns out that the that actually the quantum mechanical case corresponds to the zero entropy limit whereas the kind of General relativistic case corresponds to the infinite entropy limit but there kind of two diff two fundamentally different Notions of entropy one of which exists at the single way level one of which exists at the multi-way level and uh again the question of how these things into play is partly why we're you know why we're investigating this and it's clear that that that question has links to these quite foundational questions in fundamental physics the ideal Point Mass and the ideal distribution with it center of gravity and all this Dave question for you from you yes um okay I I was can you hear yep go for it okay good um yeah I'd like to to Veer down to the low road during this discussion uh maybe show business when people are trying to explain what is computational reducibility or irreducibility often you'll see a a graph that says well now here's the the computation our Target running along the foxes running along and behind it there's a team of algorithms that would like to catch it and yeah it either does or it doesn't outrun all of them but some can come can seem to come pretty close to catching it um now there's something else people have been looking at for a few years the mandal BR set the mandle BR set it really is a deter not only a deterministic computation it's a crisp computation every set every Point either is inside the safe Zone it's going to sit there and it's happy and quiet and it's color black or it flies off to Infinity so you could it could just be a

bunch of yes or NOS but that's not the way people trying to calm down after a a day of work want to look at they want to say hey I want to see the 32 million deep mandal BR set and I want to see it in colors so where do you get the colors you ask how hard did I have to work how long did I have to grind along before I made that decision oh this is a point that's going to settle down and be quiet and uninteresting or it's going to fly off to Infinity so you P put colors in now I just wonder would it be interesting to anyone to fuzzify these gorgeous causal and multicausal graphs and just show this is the portion where we had to work really hard rule 35 or what we had to go 15,000 Generations before we gave up and said we're not going to follow this anymore the other one oh after 30 steps I I see that's a really interesting question and and I mean yes so uh I mean a couple comments on on on that I mean so one is I mean your right in the sense that yes the M Bret is a very clean thing to describe I would I would say it's not I mean it's I would argue it's actually not a crisp computation in that sense precisely for reasons of computational irreducibility because as you go arbitrarily close to the boundary of the set you can have complex numbers that stay that have a kind of indefinite period of transient right there's there's no upper bound on how long the you know $Z^n + C$ orbit can last before it either diverges or converges to to to you know to to zero um and and and that statement that you know that there are there will be points that can you know that can remain that can get tied up in these orbits indefinitely is really a computational irreducibility statement um but yeah I mean your your question about uh you know could you could you construct so I mean that that that way of coloring points that are on the boundary of the mandel BR set in in that way that this the so-call Escape time algorithm right where where you kind of where you color them based on how many steps that I need to do before it either converged or or escaped off you know above some or you know the complex numbers modulus exceeded some value um yeah it's an interesting idea that you could try and kind of construct geometrical representations of the space of possible computations based on a kind of Escape time algorithm for computational risibility yeah so I mean there are there possible ways that you could do that right that that you you say you know we know we've known since the days of touring that the halting problem for touring machines is undecidable right there's no finite computation that you can do that will determine whether or not arbitrary computer program will will terminate in finite time so you could do a you could if you once you have a way of kind of geometri the space of possible computations which we which we have now as you say um yeah you could easily construct a kind of Escape time algorithm where you know you have your analog of the mandel BR set which is you know here's all the halting computations here

are all the definitely non-halting computations and then there's some boundary of very fuzzy stuff where we kind of don't really know and we have to do a lot of work and even then it's only heuristic um which is which is yeah I mean very directly analogous um and then yeah the question becomes you know so in in the case of the mset you know that there there is some underlying Theory mostly I think mostly due to like Dadi and hubard and people uh that allows you to kind of predict like you know if if you have a finite filament you can of the of the mandle Bret you can kind of predict you know where that where that filament is going to be based on some complicated complex analysis argument and you know Ian the question is once you if you once you have a geometrization of the space of possible computations can you construct some general theory like the theory you know developed by by D and hub that tells you things about you know the topology of the computations where the certain regions of the of the space are connected whether they're compact whether you can make pred you know whether you can do a similar thing or you can make predictions based on the geometry of one part of a set you know where you can extrapolate to things about the geometry of another part of the set that's a really interesting question um and uh yeah again it's as with so many of these things it's it's one of these like I I don't know the answer but it's a good reason for investigating you know for pursuing the program right yeah from the Wolfram side as well as from the active inference side there's both the information topology and the information geometry side and not just in the kind of topological deep learning but rather looking at the topology of information flows which has been heavily developed in the category Theory related to Quantum information sciences and then the information geometries that allow us to do like machine learning type accelerated optimization and gradients all these kinds of Concepts that come into play with geometry and we just don't get them from topology so topology kind of sketches the skeleton and then with the computers that we have the information geometry is at least on the data sets and the ways that we have computation today that's kind of like the quantitative numerical versus the formal um one question is are these all discrete State space discrete time formalisms because in active inference we often deal with Hybrid models that have discrete and continuous State spaces and the same generative model or the same system of Interest could be modeled with like a discrete Time chapter 7 or a continuous Time chapter 8 model so how does this deal with that that's a really good question I mean yeah so so so most of what I've been uh looking at so far has has consisted of um discrete time discret space uh uh models for no particularly principled reason other than um they're easier to analyze right for the most part because you can do explicit computations because they're kind of more

amable to constructive analysis um it's easier to do but the beauty of a lot of this you know that's one of the beauty of using beauties of using kind of General mathematical formalism is that once you develop it um it's often quite easy to extend even to even to cases that you know to extrapolate it to cases that you didn't explicitly analyze so um in principle this formalism works for for you know for for for Continuum space and Continuum time systems as well just with some you know with some slight modifications so rather than having say branching and merging you instead have you know if if you think about this thing as now being a dynamical system you know described on some symplectic manifold then these kind of branching merging operations of the multiway system become uh effectively Divergent and convergence oper you know differential operators defined on the simplec manifold and uh so you know one place where we can start to analyze that explicitly and which I've done a little bit of work on but it's one of these things which I I want to go back to very soon um is looking at Petry Nets so Petry Nets are interesting because they are a discrete time discrete space system but they admit a Continuum you know Continuum space continuum time description in terms of ordinary differential equations and so on so they're they're a nice example of a hybrid kind of discrete event versus Continuum event system where it's clear that this formalism can be used and and is somehow agnostic as to whether the you know whether the underlying system is discret or continuous um again there's a broader philosophical point to make here which is that in a way one of the reasons I don't feel embarrassed to be working primarily with discret systems is because you know again once you start to think about things in terms of okay you have to not just care about nature you also have to care about the computations The Observer can perform and and what it Ables to infer um then you quickly realize that in a sense just like whatever as you were saying earlier Danel you know beauty is in the eye of Beholder I think discreteness and continu and and and and continuity are also in the eye of of a holder right I mean so if you have um if you have a you know a universe that is fundamentally continuous that's described by you know a Continuum lenan manifold or something but you're constrained so that the only experiments you perform have computable outcomes have discrete outcomes uh where you know where where the countable where the the possible number of um uh you know of observables is always countable then in a sense it doesn't matter right it's it's irrelevant to you as an observer whether you know whether the whether the system is discrete or continuous uh because you know the only parts of it you can interface with and interact with are discret and so you could have replaced the underlying substrate with a purely discret mathematical structure and you wouldn't be able to tell so in some sense uh I yeah I don't feel

too embarrassed discreet dealing with discret Event Systems because even if I don't necessarily believe that nature is discreet because I don't think that's I'm not even sure how we would be able to answer that um I'm reasonably convinced that that you know the the experiments that we can perform and the observations that we're able to perform are ultimately computable and therefore you know the underlying substrate might as well be discret even if it's not sort of in reality so to speak yeah that's a great comment definitely calls back to your earlier points about like discretization being in the eye of the beholder like in the active inference models observations raw data may already be discretized depending on the situation but even if it weren't like it were a continuous sensory perception or modeled as such analytically still commonly models discretize and categorize as they move up cognitive hierarchies and that was like initially explored to get more of this discreet either or decision-making planning all those kinds of properties um well there's many interesting angles like I'm sure also it could be a multiplexed language model prompt but like what are you working on or excited about for 2024 uh that's good question um so I've kind of already given some hints about you know the like this this general research program of of of trying to understand uh computational complexity and algorithmic complexity and interplays between observers and systems through this category theoretic lens that's a yeah that's a that's a major thing which I started on I say about a maybe a couple years ago I mean in some form I was I've been working on it for a long time but but this more recent perspective on it is maybe a couple years old but for various reasons over the last year or so I've kind of put that to rest and I've been focused on these uh much more physics oriented questions about discrete SpaceTime and understanding things like you know how do black holes work and how does secretion work in discrete SpaceTime which is also very important and uh and and very exciting but uh I've sort of slightly been missing these more abstract directions and so I have maybe one or two major you know physics related things that I need to finish off and then I and then I really want to go back to this to to to the greatest extent possible and yeah I mean so one thing is that's quite clear is that there's uh um there's great interplay between this formalism and ex you know an existing theories of computational and algorithmic complexity so in particular you know so uh one very basic example is I mentioned before that you know you you you have this kind of coherence between these two different algebraic structures between your the the operation of Tim like composition versus the operation of kind of parallel composition and these two algebraic structures are in general related although the precise conditions that relate them are not clear and that's partly what we're trying to what we're trying to understand um but

it turns out that that uh degenerate cases of that of that question corresponds to unsolved problems in computational complexity Theory so for instance the P vs NP problem can essentially be recast in these terms that you can recast the P vs NP problem is the question about uh is the coherence between the timelike composition of computational complexity and the parallel composition of computational complexity which are what P and NP respectively are really about are those coherence conditions the the the strictest they can be um which would be the case that that the P equals NP or are they somehow more LAX which would be the case that P does not equal NP and so there's you know that's that's one thing that we kind of already investigated but it's clear that a whole bunch of questions about you know um how the time complexity and space complexity trade off or how do log of complexity and time complexity tradeoff these these are questions which can be recast in this kind of more algebraic category through lens and and and um will hopefully give insight into this general program of trying to understand observers and and their relationship to the world and those are kind of major um well with any luck those are major theorems that I hope we'll be able to prove at some point in 2024 um that's but that's sorry that's that's awesome and it makes me think about parallel more nestmates more CPU threads deeper in time more sequential more planning and more cognitive single monolithic agent and then the kind of question is like can anything that a single agent Mega Matrix could do cannot decomposable at space advantage or even at space disadvantage in decomposed it into a single time step operation yeah that's a that's a super important question and one that you know uh with the possible exception of this community not many people have asked right I mean so uh that's something which comes up in Quantum Computing right so um a lot of the hype around Quantum computation comes from these theoretical speedups that derive from the fact that you're able to uh you know you're able to support these superp positions of different you know where you know each each state of your data structure corresponds to a different IG State and you're able to evolve some super superposition of those igen States but then at the end you have to actually come to a definite conclusion about what the answer is you have to perform some measurement operation and that measurement operation is lossy it's often non-deterministic uh you you know often have to repeat it mult multiple times um and you know it's it's becoming increasingly clear that for a large class of operations that were previously thought to have Quantum Advantage the additional complexity of the measurement step really kills any Quantum advantage that you may have had that that you you know you get some Advantage by doing unitary Evolution uh but then you lose all of it by having to do this submission projection at the end and that's really

a story of again this interplay between the time complexity saving of doing a multi doing a computation multi doing a multi computation in parallel versus the loss that comes from the the complexity of the equivalence function that you need to apply in order to get to some definite uh conclusion about what the you know about what happened because you know ultimately you need to somehow collapse that that that directed graph into a single thread uh of time in order to be able to have some coherent representation of what happened and so again understand that you know that's a place where understanding these you know these trade-offs will become very important and as I say in some limit in case that that gives some perspective on your question Daniel right which is which you know I agree is a very interesting question question about you know um in principle we know that anything that a deterministic touring machine can do a non-deterministic touring machine can do and vice versa with some speed up or slowdown um but that statement which is a classic result in you know in computability theory neglects all consideration of the equivalence function so there may be cases where the equivalence function is so complex that essentially the you know the the to do state equivalence becomes undecidable and so in that case you have a scenario where actually you you know you've got a multi computational system but to to collapse it to one that's equivalent to a single way system requires unbounded amounts of computational effort and so so actually they become inequivalent even though you know computability Theory says they should be the same um so uh it's clear that there's a there's a a more Rich more subtle Theory that's underlying here that we're just beginning to kind of Glimpse and that I hope we'll you know we'll be able to kind of to to prove some new limited results about soon once we understand it a bit better awesome and I think definitely a special shout out to all of our colleagues on either like the wol from Andor octave inference side because we've seen few if any active inference models phrased analytically or computationally with the wol from technology um from studying complexity and other areas it's really clear to see how productive and Powerful the software and the tools can be and changing and growing every day it's really interesting maybe someone can um if they're listening this far in like go for one side to the other and back or make a wol from active inference model or do some other kind of combination because it's very fruitful territory and we know that our um Elders have already spoken they vocate it no but really it's so many it's so rich with connections here between um the areas that we're all studying and feeling like converging on many common places to to scaffold and jump off from together yeah I I I I agree and um I mean at least the I mean the kinds of things that we were discussing here about you know speculative execution and and

and behavior formation through free energy principle and so on those things should be relatively easy to implement in the framework that's already been developed here I mean so so you know that that's that's just a you know that's a question of just implementing some kind of computation of expected free energy and using that to you know to to to to to wait multiway paths in the speculative execution model um so at least the beginnings of of that implementation I think the the path is is pretty clear and we will probably end up doing at some point in the future anyway uh as part of as part of other research um so yeah so I agree that's it's a very it's a very exciting kind of um point of interface yeah well I hope that that um we can stay in touch if you ever want to come back for a uh 9.2 or if we want to even facilitate some kind of working group or some connections to really strengthen and like include the participation of more people in this super exciting area that would be amazing yeah that sounds fun let's let's try and set thing up well Dave first penultimate comments then Jonathan you can have the kind of last comments yes um I hope you do get to continue on the wolf physics side to think about a more General notion of what these ultimate things are uh are they observers or does that already Prejudice the case of what you might find if you call them workers or actors or you know go back and and ask steuart Kaufman what must a mind do to earn its way in the world let's make this keep happening thank you thank you Dave for suggesting it also it was a great suggestion Jonathan no that's I think that's a fantastic note to end on I mean and and that that in a sense this this idea that we should start to move I mean so it's okay big picture for a moment like we're you know this formalism is being developed the the the formalism I've described in this in this discussion is being developed you know assuming a kind of purely passive Observer idealization and that's already been incredibly difficult right it's clear there's a lot we don't understand at that but of course David is right that that that in a sense you know what you know ultimately we want to start transitioning to a to a participatory Observer model where where you allow for two-way interactions or you know higher order interactions between observers and systems and things that don't just go in One Direction and yeah you know in a sense I view a lot of what we're trying to do with you know trying to nail down these Notions of causality trying to understand these interplays between different complexity and entropy measures as you know the necessary groundwork for developing that subsequent Theory right that that you know it's clear that if we want to have a a version of the you know of this kind of compositional multi-way formalism that is also compatible with things like second order cybernetics then uh you know at the very least we need to have a very coherent notion for what causality is and and and you know and a robust

algebraic description of that that's not going to to to break or change uh and so I think um the the non-participatory participatory Observer model is a useful starting point because it's one that is just within our grasp of uh you know of being being kind of mathematically tractable um and then the hope is that the technology and the the ideas and and the conceptual structure that we develop for understanding that will then as I say lay the groundwork for for developing something that's that's more like what real you know active participatory observers do um and uh and yeah I mean I I I think I I don't on the on the scientific side I'm not really sure I have anything any final comments apart from just you know as is obvious this is a you know this is still a story that's being that's being developed and and yeah as Daniel alluded to I hope that we can continue to uh to interact and and collaborate where where that makes sense and and yeah at the very least I think incorporation of um kind of these you these active inference models Within These uh discrete time in in the first instance discrete time uh you know computational Frameworks uh and allowing for things like speculative execution and multi-way path waiting based on um free energy estimates uh I think that's you know that that that's a um that's a project that's obvious mutual interest and and and something that I that I that I hope will happen in the coming months thank you we just speculatively executed AC of wolf from reference basically s all right thank you Jonathan thank you Dave thank you everyone see you all next time